R13

Code No: 126AQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, April - 2018
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Used paper 23-04-2018

PART - A

(25 Marks)

1.a)	Explain the Caesar cipher.	[2]
b)	Define confidentiality and authentication	[3]
c)	Differentiate conventional & public key encryption.	[2]
d)	What is traffic Padding? What is its purpose?	[3]
e)	What is the purpose of X.509 standard?	[2]
f)	In the content of Kerberos, what is realm?	[3]
g)	Explain the reasons for using PGP.	[2]
h)	Compare Transport mode and Tunnel Mode.	[3]
i)	What do you mean by malicious software?	[2]
j)	What is application level gateway?	[3]

PART - B

(50 Marks)

2.a) Explain how gateway works in internetwork security model.
b) Explain the various types of cryptanalytic attacks. [5+5]

OR

3. Explain symmetric and asymmetric key criptography. [10]

4.a) Explain how key exchange is done using Diffie-Hellman key exchange.
b) Discuss the "man-in-the-middle" attack. [5+5]

OR

5. Explain Blowfish algorithm. [10]

6.a) What is HMAC and what are its advantages over MAC?
b) Discuss different approaches to Message Authentication. [5+5]

OR

7. Discuss the requirements of Kerberos. Explain the Kerberos ver-4 message exchanges. [10]

8.a) Explain the different MIME content types.
b) Explain S/MIME certificate processing method. [5+5]

OR

9.a) What are the applications of IP security?
b) Describe the general structure of IPSEC authentication header. Discuss how anti-reply service is supported. [5+5]

10. Explain the different types of firewalls with neat diagrams. [10]

OR

11. Discuss about Intrusion Detection and approaches of Intrusion Detection. [10]

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